

Total No. of Questions : 10]

SEAT No. :

PA-331

[Total No. of Pages : 4

[5927]-215

B.E. (Mechanical)

REFRIGERATION AND AIR CONDITIONING
(2015 Pattern) (Semester - I) (302049)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Solve Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8, Q.9 or Q.10.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary and mention it clearly.
- 5) Use of steam table is allowed.

Q1) a) Explain with neat sketch Bell-Coleman Cycle. Also represent the cycle on P-V and T-s diagram. **[6]**

b) Explain the working of three fluid vapour absorption system with neat sketch. **[4]**

OR

Q2) a) State desirable properties of a good refrigerant. **[4]**

b) A vapour compression system having capacity of 4 TR and uses R12 as a refrigerant works on simple saturated cycle with dry compression between evaporator temperature of – 15°C & condenser temperature of 30°C. Take specific heat of vapour refrigerant as 0.62 kJ/kgK. Represent the cycle on p-h and T-s diagram and Calculate : **[6]**

- i) Mass flow rate of refrigerant circulated through the system.
- ii) Power required to drive the compressor.
- iii) COP of the system.
- iv) Heat rejected in condenser.

Properties of R12 are :

T(°C)	h_f (kJ/kg)	h_g (kJ/kg)	s_f (kJ/kgK)	s_g (kJ/kgK)
–15	22.33	180.98	0.0906	0.7052
30	64.59	199.62	0.24	0.6853

P.T.O.

- Q3)** a) Explain water cooler with neat sketch. [4]
b) Explain with neat sketch the working of two stage compression with water intercooler and liquid subcooler employed for vapour compression system. Represent the cycle on p-h diagram. [6]

OR

- Q4)** A 40 TR two stage compression ammonia refrigeration system uses water intercooler between two stages and liquid subcooler for undercooling of refrigerant leaving the condenser. The system is working between the evaporator temperature of -20°C and condenser temperature of 40°C . The intercooler pressure is 6 bar. The ammonia is cooled to 30°C in water intercooler and subcooled as liquid to 30°C . [10]

Find :

- a) The mass flow rate of ammonia circulation
b) Power required to drive the system and
c) COP of the system. Also represent the cycle on p-h diagram. [Use Ammonia (R717) refrigerant chart]
- Q5)** a) Define specific humidity, relative humidity, dew point temperature and degree of saturation. [8]
b) Without using psychrometric chart, calculate specific humidity, relative humidity, humid specific heat and enthalpy of moist air when DBT is 30°C , DPT is 15°C and total barometric pressure is 101.325 kPa. [8]

OR

- Q6)** a) Moist air at 27°C and 80% RH is adiabatically dehumidified by passing through a drier so that its final relative humidity is 30%. The air is then cooled by passing through cooler until its temperature reaches to 25°C without change in specific humidity. Show the process on psychrometric chart and Determine [10]
[Use of Psychrometric Chart is allowed]
i) Temperature of air leaving the drier.
ii) DPT of the air leaving the drier.
iii) Heat rejected during the cooling process.
iv) Relative humidity at the end of cooling process.
v) The moisture removed during dehumidification process.
- b) Explain Adiabatic Mixing of Two Air Streams. [6]

- Q7)** a) Give the classification of refrigerant compressors. [2]
b) Explain Flooded Evaporator with neat sketch. [8]
c) Explain Winter air conditioning system with neat sketch. [8]

OR

- Q8) a)** Enlist the different expansion devices used in refrigeration and air conditioning systems. [2]
- b)** Classify the condensers and explain evaporative condenser with neat sketch. [8]
- c)** Explain Thermostatic Expansion Valve with neat sketch. [8]

Q9) a) What are the different methods of duct design? Explain any one method in detail. [8]

- b)** A rectangular duct $0.8\text{m} \times 0.5\text{m}$ size carries $4\text{m}^3/\text{s}$ of air having density of 1.2 kg/m^3 . Determine the equivalent diameter of the circular duct if [8]

- i) the quantity of air carried is same
ii) velocity of air is same

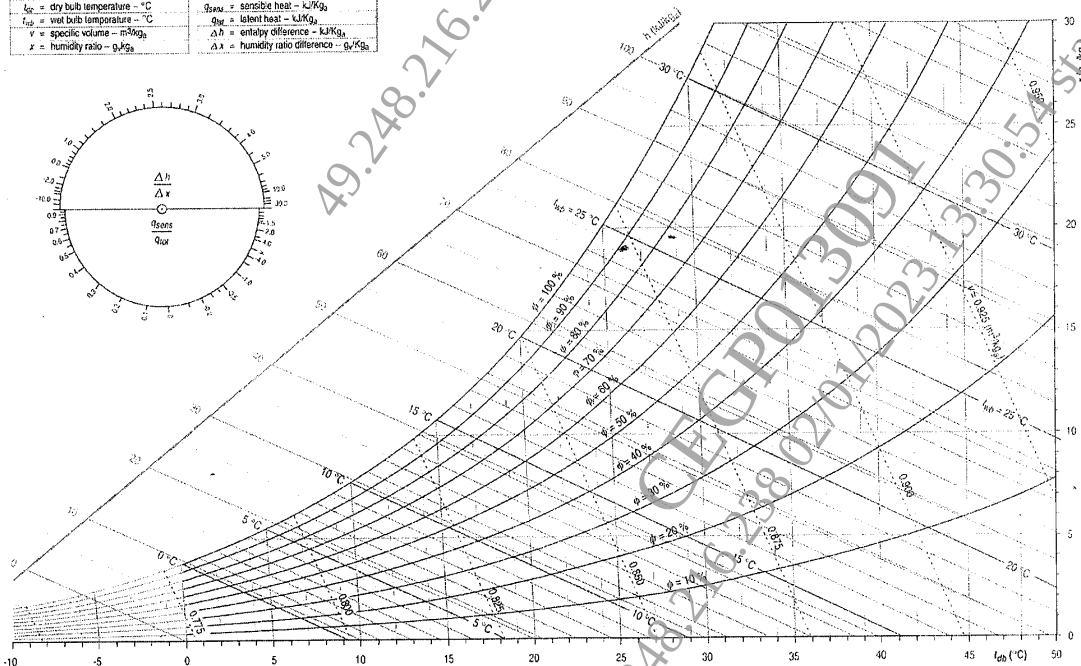
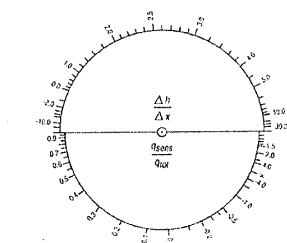
If friction factor $f = 0.001$, Find the pressure loss per 100 m length of duct

OR

Q10) a) Derive the expression for equivalent diameter of circular duct corresponding to a rectangular duct of sides 'a' and 'b' for the same pressure loss per unit length, when the discharge is same and when the velocity of air flowing through both ducts is same. [10]

- b)** A rectangular duct $400\text{mm} \times 200\text{mm}$ is 20m long and carries $30\text{m}^3/\text{min}$, of air having density 1.2kg/m^3 . Find the total pressure required at the inlet to the duct and air power required. Take friction factor $f = 0.005$. [6]

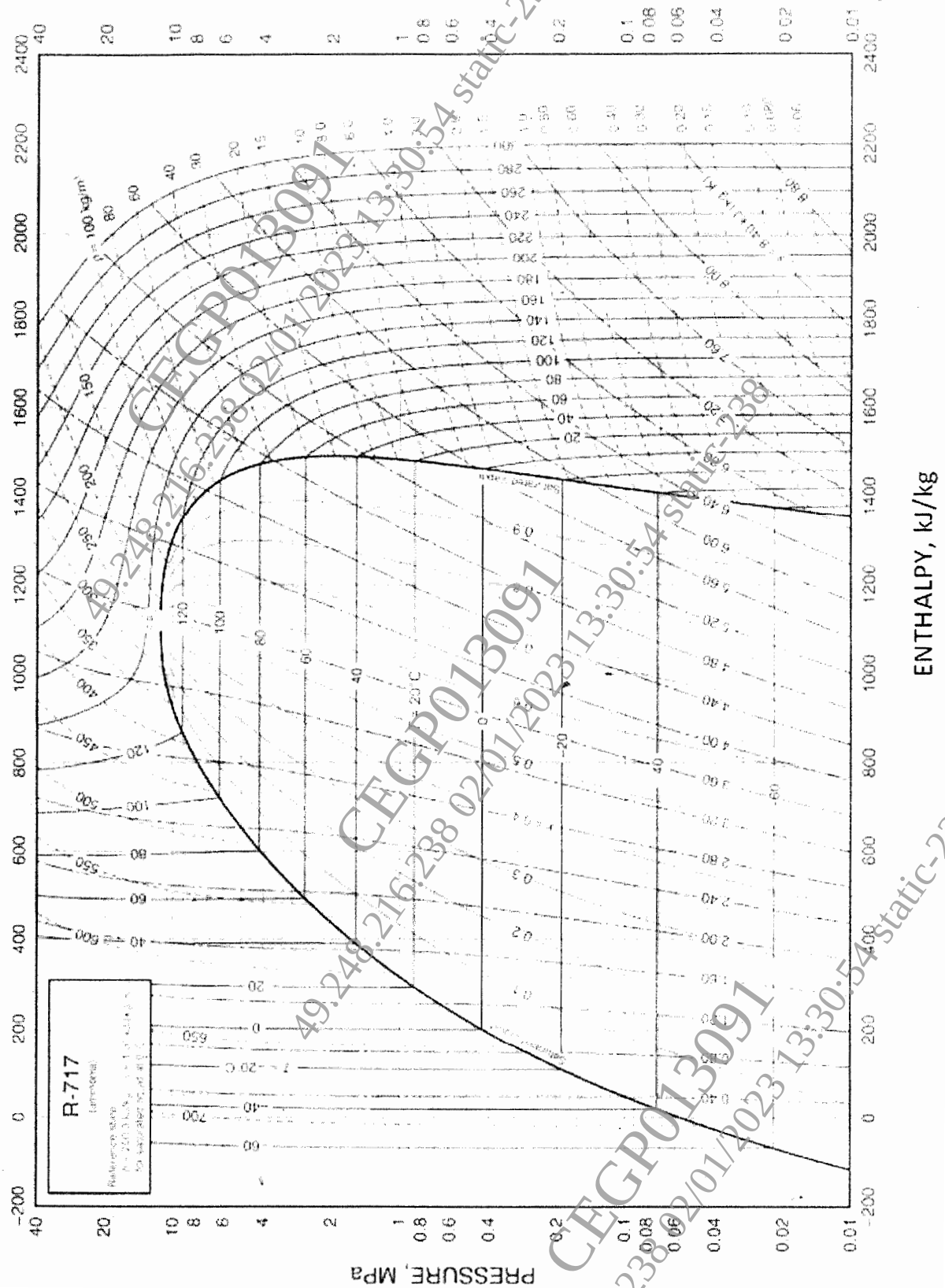
t = wet-bulb temperature - °C	ϕ = relative humidity - ad-h
t_{dp} = dry-bulb temperature - °C	q_{sens} = sensible heat - kJ/kg
v = specific volume - m ³ /kg	q_{lat} = latent heat - kJ/kg
x = humidity ratio - g/kg	Δh = enthalpy difference - kJ/kg
	Δx = humidity ratio difference - g/kg



x

x

x



ENTHALPY, kJ/kg

49.248.216.238 02/01/2023 13:30:54 Static-238